

High-level tools for galaxy simulations analysis

Dr Kyle Oman

Royal Society Dorothy Hodgkin Fellow

Institute for Computational Cosmology

Durham University



Optimal data storage & access is hard

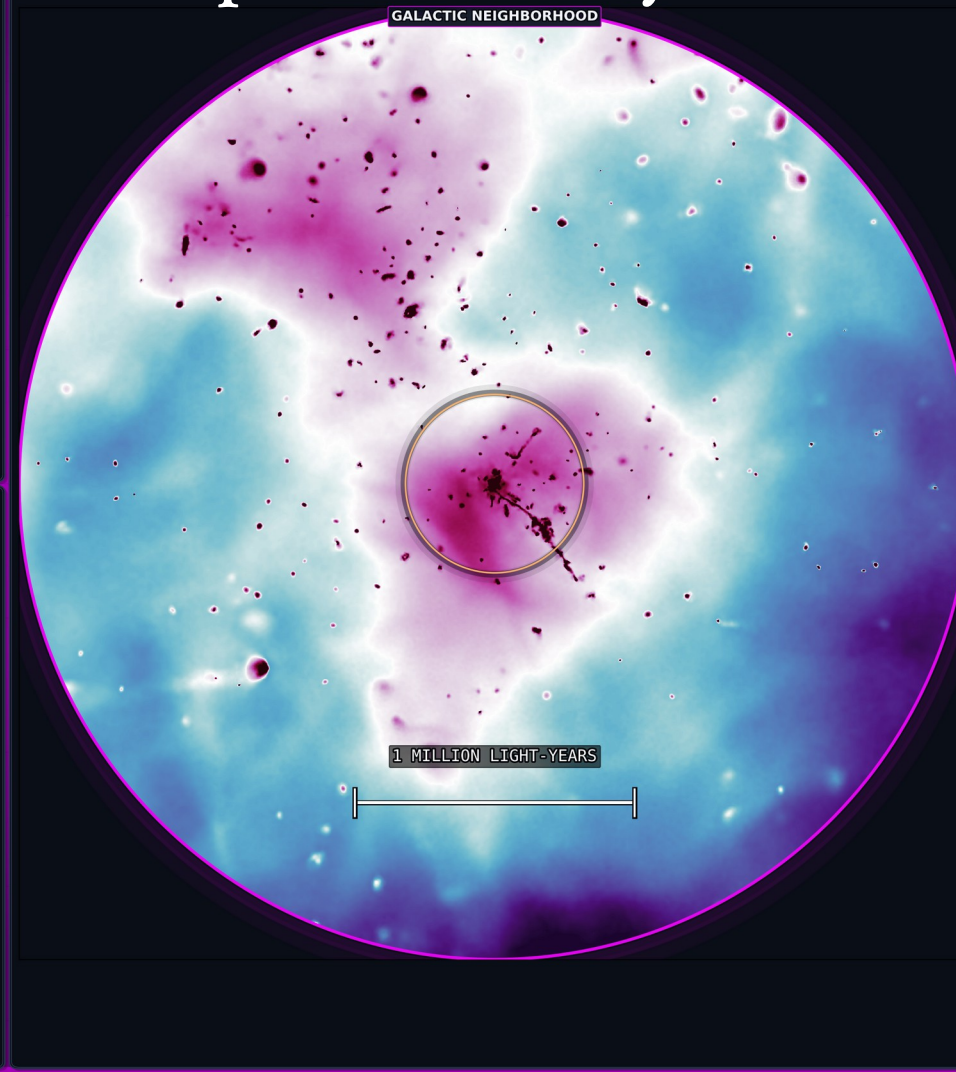
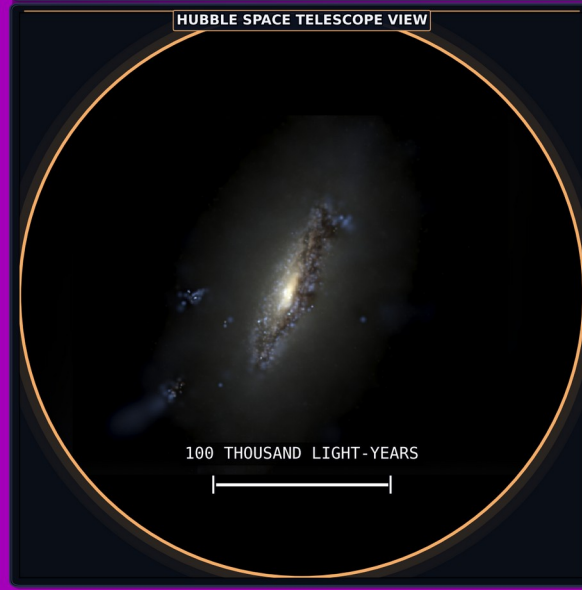
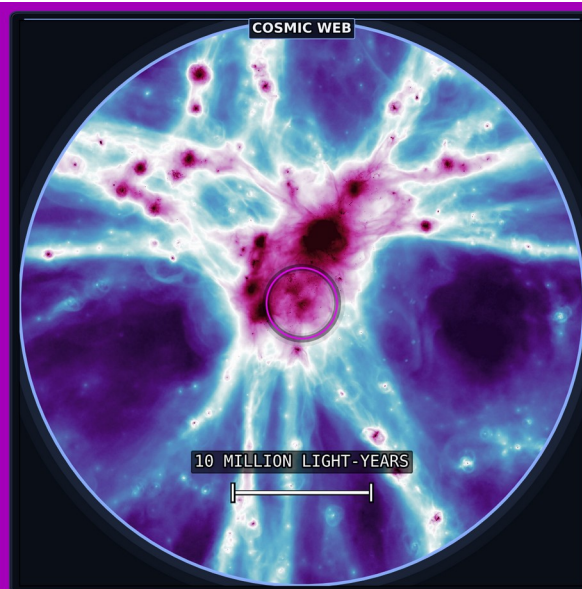
- Enormous dynamic range.
- Many use cases and access patterns.
 - Anything from $1/10,000,000^{\text{th}}$ of the data to “all of it”.

Example 1: Cosmology

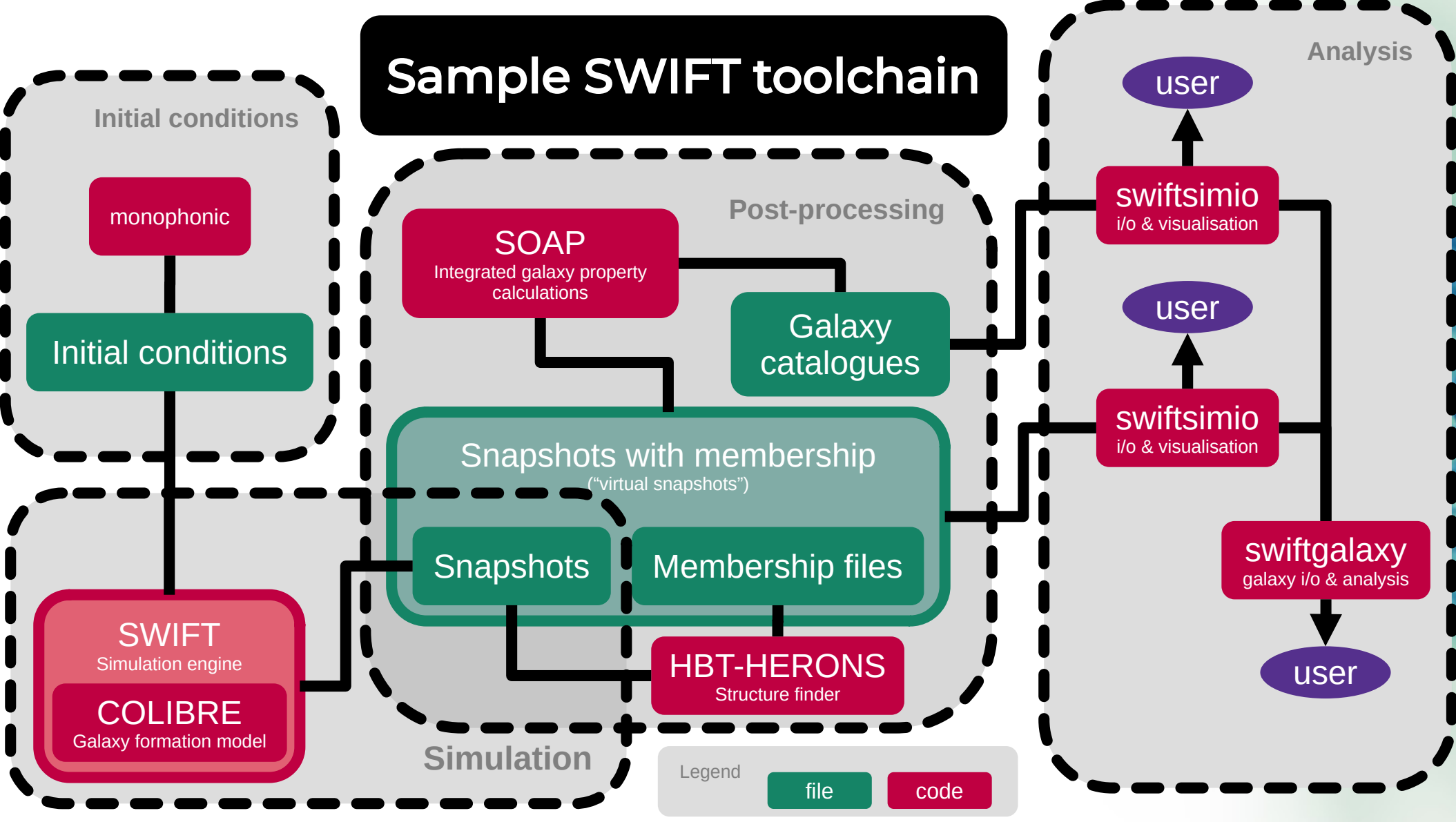


2.8 Gpc

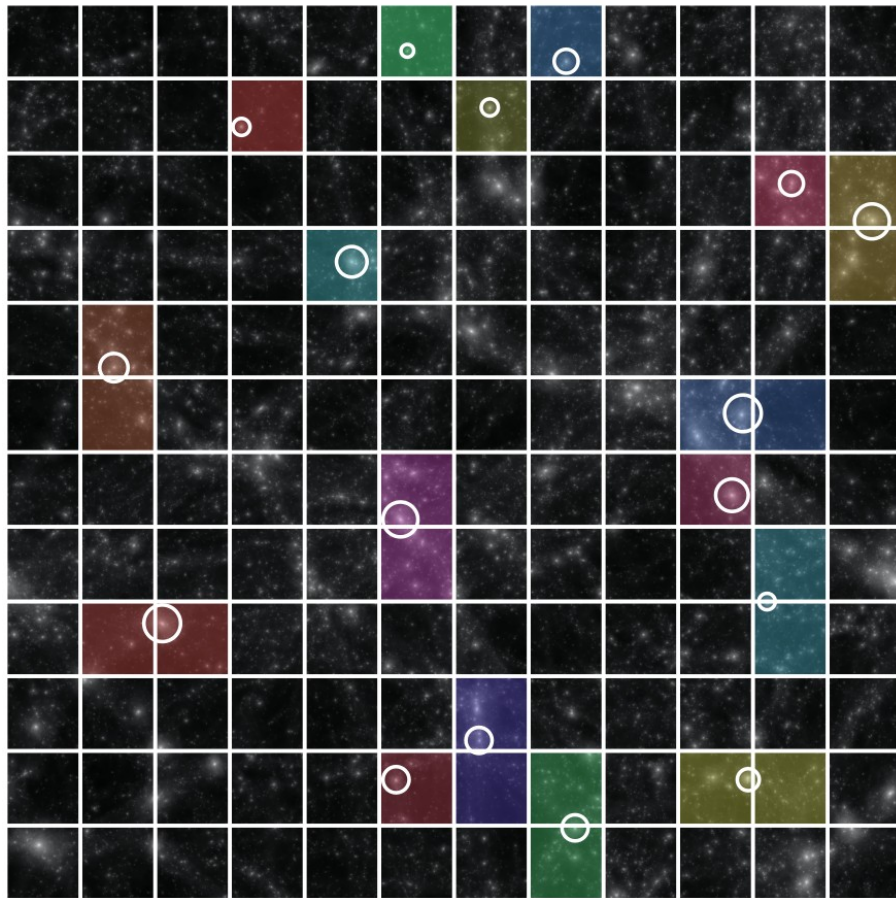
Example 2: Galaxy



Sample SWIFT toolchain



Selective data loading



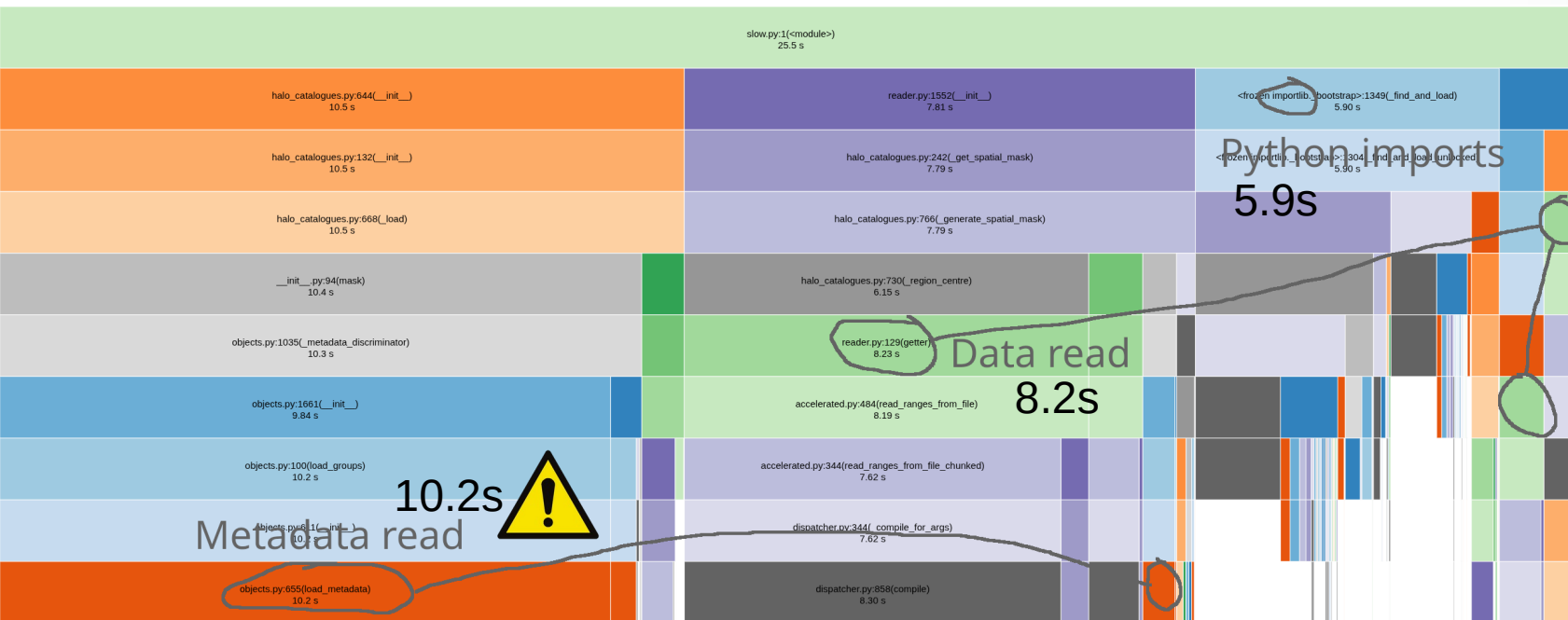
- Relies on:
 - spatially structured data files
 - metadata describing structure
- Limited to:
 - workflows needing spatial selection
 - cuboid(-ish) regions

This slide illustrates lazy-loading

- I/O is only triggered when data are required.
- If more data are required later, another I/O task is triggered.

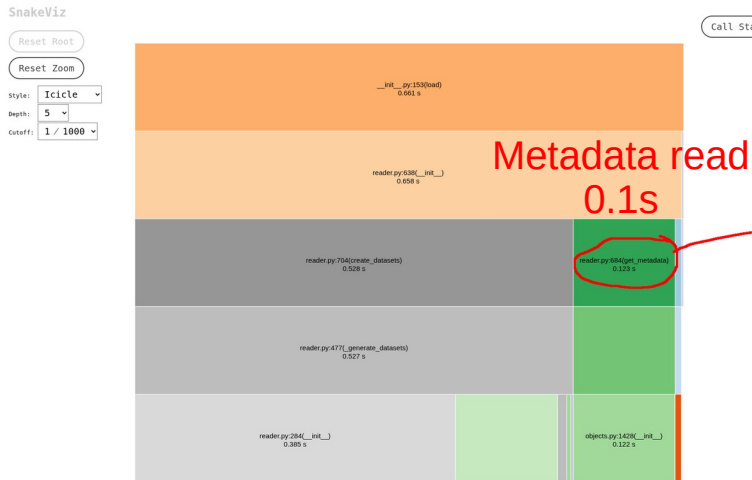
~~Metadata access is too slow~~ Data access is too fast

Profile: old version

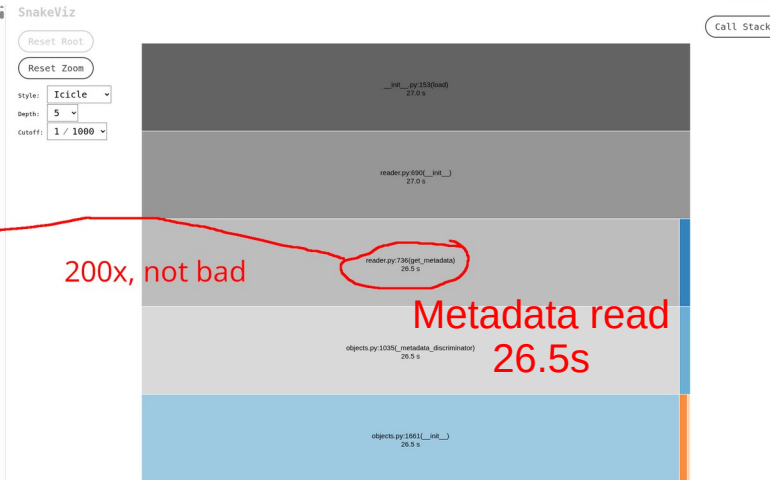


Solution: lazy-load metadata, too

Profile: optimized



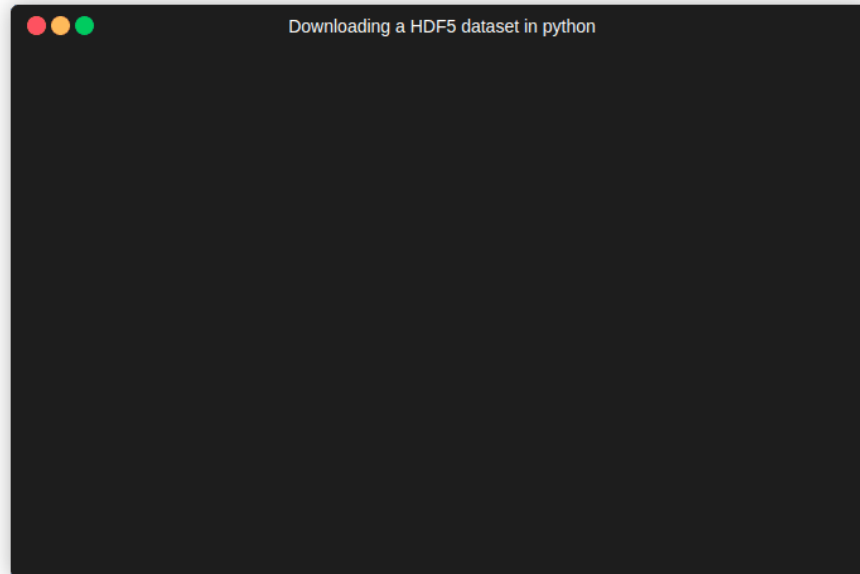
Profile: old version

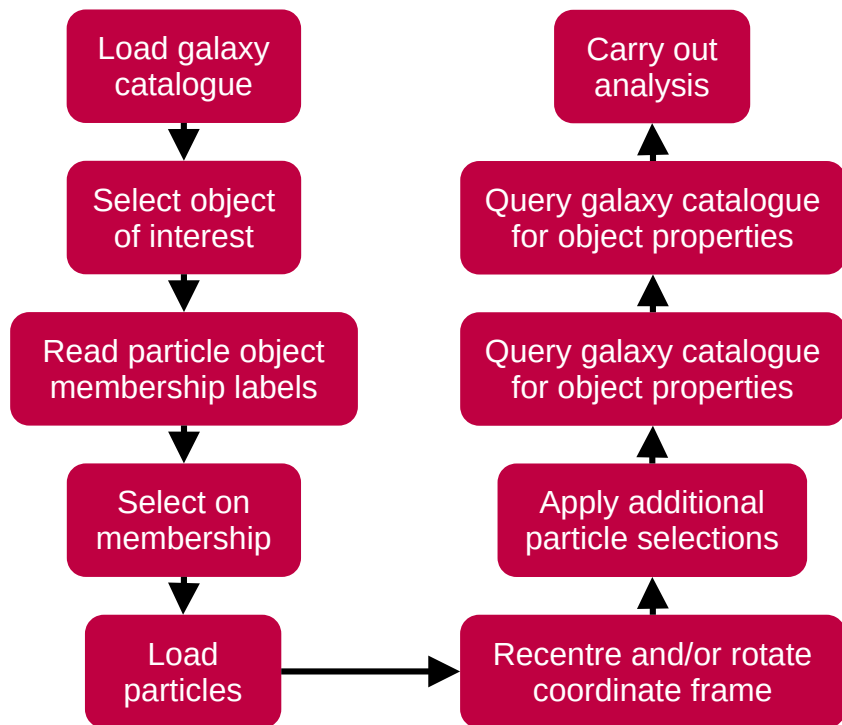
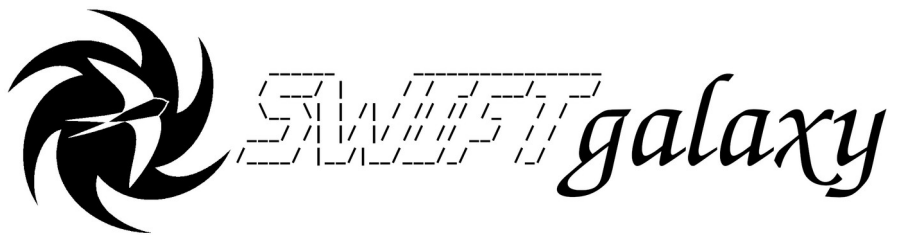


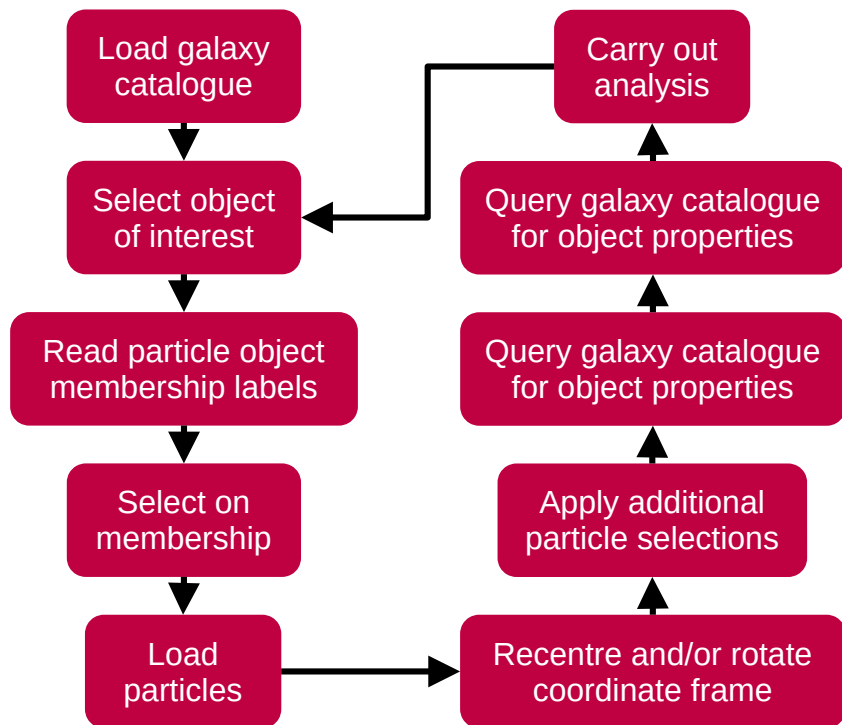
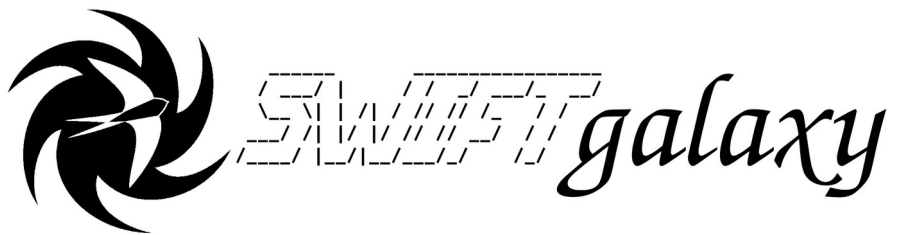
SWIFTSimIO features III:

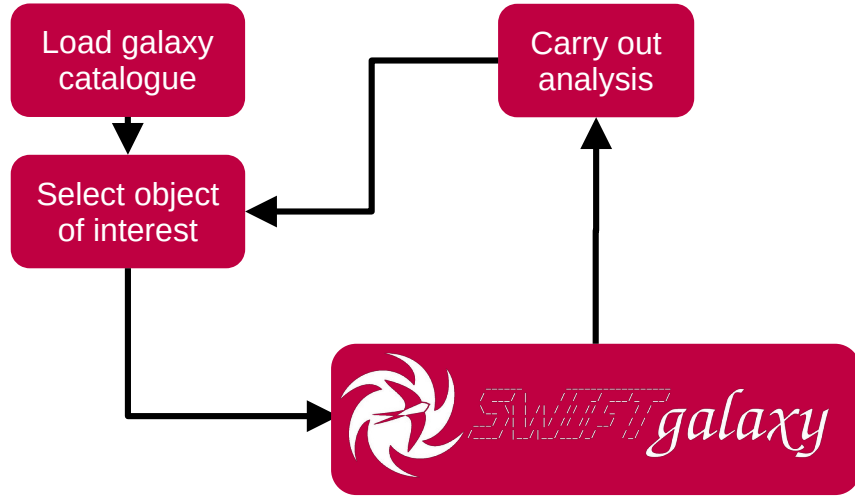
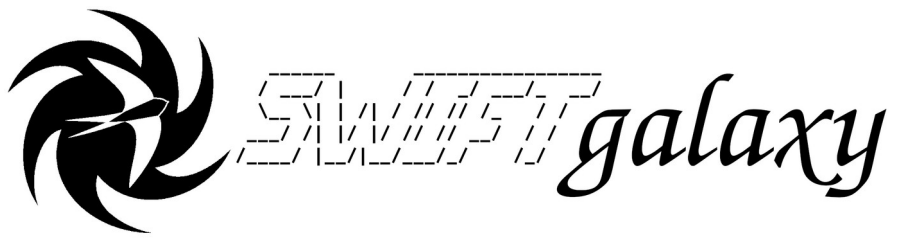
Accessibility

- Many more people have a laptop than have a compute cluster

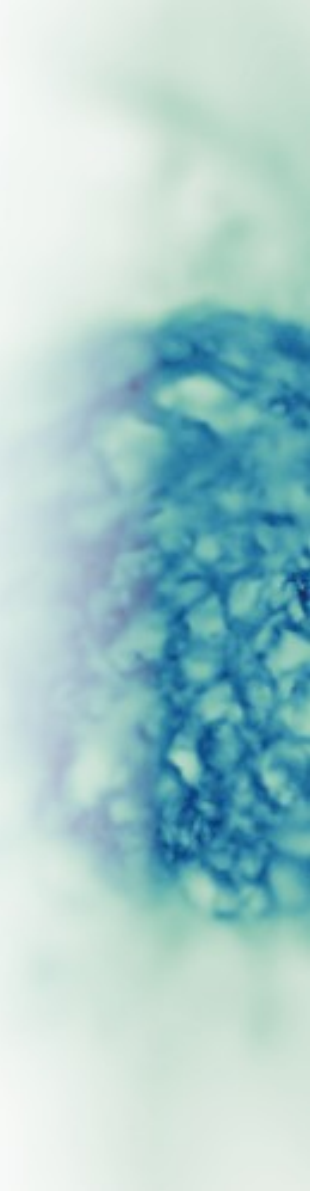
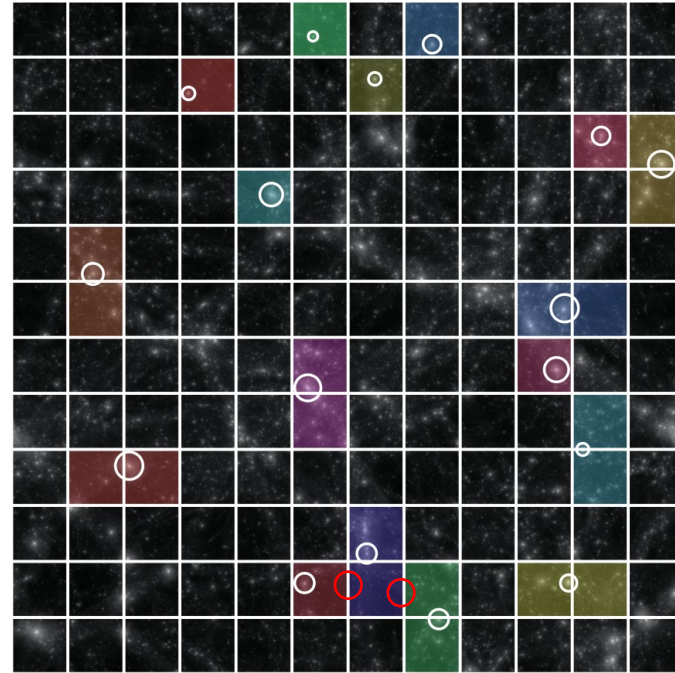








```
>>> SWIFTGalaxy(  
...     "data_file.hdf5",  
...     SOAP(  
...         "catalogue_file.hdf5",  
...         object_identifier,  
...     )  
... )
```



Other considerations

- Compression.
- Data integrity & redundancy.
- Balancing performance & user skills.
- Role of relational databases.
- Parallel I/O.

Summary

- Optimal data storage and access for analysis is an ongoing process.
- Involves how data are produced, stored and tools to read them.
- Lots of tensions in optimization.